Trevor Henry Chiboora

Security Engineer | Security Monitoring & EDR | Penetration Testing | Data Protection

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Relocating to Kigali, Rwanda, in December 2026. Open to on-site and remote roles. No work sponsorship required.

Summary

Security engineer with about three years in security operations, penetration testing and security research at the Upanzi Network, Carnegie Mellon University Africa (Kigali), and Approov (UK). Before that, five years in IT operations and teaching. I have run full-cycle penetration tests, from testing through retest; monitored 15+ servers with SIEM, IDS and EDR tooling; and operated internet-facing honeypots for a year. First author of an ACM paper on mobile banking security and co-author of data protection research across 49 African countries. Individual AfricaCERT award recipient. Completing an MS in Cybersecurity & Privacy at NJIT (December 2026).

Skills

SOC, EDR and IR: alert triage, threat hunting, escalation, EDR host isolation, runbooks; Elastic Security, Elastic Defend, Wazuh, Security Onion, Suricata, MISP

Penetration testing: web apps, APIs and Linux servers; Burp Suite, OWASP ZAP, Nmap, Masscan, OpenVAS, Nuclei, Nikto, Gobuster, sqlmap, Metasploit, Kali

Mobile and malware analysis: OWASP MASVS, MobSF, jadx, apktool, Frida, Objection; CyberChef, Ghidra, VirusTotal, ANY.RUN; honeypots (Cowrie, OpenCanary)

Infrastructure and access: firewall rules, VPN, network segmentation, backups and recovery, Linux hardening, Active Directory, Windows Server, SSH key and sudo management, access reviews

Data protection: Rwanda Law N° 058/2021, privacy-by-design, consent and privacy-notice analysis, compliance gap assessment

Detection and AI security: Wazuh rule writing, MITRE ATT&CK mapping, prompt-injection testing, OWASP Top 10 for LLM Applications

Scripting: Python, Bash, pytest, GitHub Actions

Experience

Cybersecurity Research Engineer Jan 2025 – Aug 2025
The Upanzi Network, Carnegie Mellon University Africa, Kigali, Rwanda

- Ran full-cycle penetration tests on web applications, APIs and Linux servers, including externally facing apps: testing, formal findings reports, walkthroughs with system owners, remediation tracking to closure, and retesting. Typical findings: misconfigurations, default credentials, outdated vulnerable software, IDOR.
- Designed, deployed, operated and analysed internet-facing honeypots (Cowrie, OpenCanary) for the MIMON and ATCHEDJI projects over about a year. Captured malware samples, attacker commands, credentials and source data; fed indicators to ELK dashboards and MISP. Received an individual AfricaCERT Recognition and Appreciation Award (2024).
- Deobfuscated attacker commands and analysed captured malware with CyberChef, online sandboxes and Ghidra.
- Automated server hardening and rootkit scanning (applied to every new server and before each penetration test), VAPT reconnaissance, log collection and agent deployment in Python and Bash. Wrote runbooks for deployment, updates and hardening.
- Mentored three interns and student project groups.

Research Associate Jan 2023 – Dec 2024
The Upanzi Network, Carnegie Mellon University Africa, Kigali, Rwanda

- Monitored 15+ servers across research, development and monitoring projects with Elastic Security, Security Onion and Suricata. Triaged high alert volumes, ran threat hunts, and escalated findings to project leads.
- Deployed and maintained Elastic Agent with Elastic Defend EDR and Wazuh agents. Responded to endpoint alerts, including isolating a host on a malicious-download detection and releasing it after verifying the file was an authorised security tool.
- Managed SSH keys, sudo rights and access reviews. Configured firewall rules, VPN access, network segmentation and server backups.
- Integrated MISP threat intelligence with the ELK stack and Suricata to enrich alerts.
- Co-authored a study of 224 financial Android apps from across Africa: 95% leaked secrets such as API keys, signing keys and payment-gateway credentials.
- First author of a peer-reviewed ACM paper assessing 18 African mobile banking apps against OWASP MASVS; presented it as an invited talk at the University of Washington.

Mobile Application Security Research Intern Jun 2023 – Sep 2023
Approov Mobile Security, United Kingdom (remote)

- Tested financial Android apps against OWASP MASVS using static analysis (MobSF, jadx, apktool) and dynamic analysis (Frida, Objection, SSL-pinning bypass).
- Co-wrote an industry report on mobile app security in Africa and presented findings in live webinars, working remotely across time zones.

Cybersecurity Intern Jun 2022 – Aug 2022
The Upanzi Network, Carnegie Mellon University Africa, Kigali, Rwanda

- Ran security scans and tests, and set up lab environments and tooling.

IT Support Specialist Sep 2021 – Jun 2022
Carnegie Mellon University Africa, Kigali, Rwanda

- Supported 100+ staff and students across hardware, software, network and AV, and wrote troubleshooting guides.

Teaching Assistant Mar 2019 – Jun 2021
Harare Institute of Technology, Zimbabwe

- Taught labs and tutorials for 70+ students per term and supervised student projects; two teams earned Distinctions.

ICT Officer (ICT Intern, Aug 2015 – Aug 2016) Sep 2017 – Jul 2018
Zimasco (Pvt) Ltd, Kwekwe, Zimbabwe

- Provided Tier II IT and security support to 500+ users. Administered Active Directory, Windows Server, LAN/WAN and backups.
- Maintained endpoint antivirus protection and configured email filtering rules.
- Supported internal IT audits: gathered evidence, fixed findings and checked compliance with IT policies.

Projects

SOC homelab and AI triage agent 2026
github.com/tchiboor/cybersecurity-soc-homelab | write-ups at chiboorat.medium.com

- Built a segmented attack-and-detect lab (Proxmox, OPNsense, Suricata, Wazuh, Kali). Wrote a Wazuh rule that flags a successful SSH login after repeated failures, validated against real logs and mapped to MITRE ATT&CK.
- Built an offline LLM alert-triage agent with severity computed in code, then attacked it with prompt injection and OWASP LLM test cases. Fixed the two gaps the tests exposed and added CI regression tests.

Independent data protection work 2026

- Designing and running a measurement study of Rwandan websites against Law N° 058/2021: passive browser-based collection that never accepts cookies, privacy notices coded with local AI models checked against human coders, ethics grounded in the Menlo Report. In progress; preprint expected early 2027.
- Drafted a 17-question scored Law N° 058/2021 readiness self-assessment, now in pilot, and built a compliance service kit (gap assessment checklist, risk register, report and privacy notice templates).

Publications and talks

- **Chiboora, T. H.**, Chacha, L., Byagutangaza, T., Gueye, A. Evaluating Mobile Banking Application Security Posture Using the OWASP's MASVS Framework. *ACM COMPASS '23*, pp. 99–106. doi.org/10.1145/3588001.3609367
- Musiime, J., Gueye, A., Atombo-Sackey, N. A., Chacha, L., **Chiboora, T.**, et al. Cookie Consent and Digital Privacy in Africa: An Analysis of Website Practices and Existing Regulatory Gaps. *ACM COMPASS 2025*.
- Byagutangaza, T., Chacha, L., **Chiboora, T. H.**, Musiime, J. J., McGregor, G. The Security Challenges of Financial Mobile Apps in Africa. *Approov / CyLab-Africa report*, 2023.
- Invited talk, University of Washington Change Seminar, February 2024.

Education

- MS Cybersecurity & Privacy, New Jersey Institute of Technology (expected Dec 2026)
- MS Information Technology, Carnegie Mellon University Africa (2023)
- BTech (Hons) Information Technology, First Class, Harare Institute of Technology (2017)

Certifications and awards

- CompTIA PenTest+, valid to Dec 2027
- Cisco CCNA Cybersecurity (CyberOps Associate), expired
- ISC2 Certified in Cybersecurity (CC), expired
- AfricaCERT Recognition and Appreciation Award, 2024

Languages: English (professional), Shona (native)